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RECONFIGURABLE THERMALLY ACTUATED TRANSMISSIVE AND REFLECTIVE METASURFACES

Abstract

A reconfigurable thermally actuated transmissive and reflective panel or surface is disclosed. A panel includes a metallization layer that includes unit cells configured to resonate at a desired frequency or range of frequencies. The panel includes a switch layer formed of a metal insulator transition material that is thermally controlled with a refractory heater layer to be in a metallic state or an insulator state. The state determines whether the panel operates in a reflective mode in which incident signals are reflected or a transmission mode in which incident signals are transmitted through the panel. The refractory heater layer, when provided with a voltage, generates heat to heat the switch layer and switch to the metallic state. Removing (or lowering) the voltage allows the temperature of the switch layer to drop until the switch layer transitions to the insulation state and the panel operates in the transmission mode.

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Background/Summary

FIELD OF THE INVENTION

[0001] Embodiments of the present invention generally relate to reconfigurable intelligent surfaces. More particularly, at least some embodiments of the invention relate to systems, hardware, software, computer-readable media, and methods for reconfigurable surfaces for reflecting and/or transmitting incident signals.

BACKGROUND

[0002] Cellular wireless communication is primarily based on 5G technology, which is generally intended to deliver Gigabit data speeds, low latencies, and a better user experience. These goals have been largely achieved. The telecom industry and others, however, are working to deliver sixth generation (6G) wireless communications and networks.

[0003] Advancing to 6G wireless communication will have rigorous requirements, including extraordinarily high spectrum and energy efficiency, microsecond latency, and full-dimensional network coverage. Current communication technologies, such as ultra-massive multiple-input multiple-output (UM-MIMO) and ultra-dense networks (UDNs), are being expanded to meet these objectives.

[0004] However, these enhancements are not without their challenges. The escalation in the quantity of antennas/base stations and usage of high carrier frequencies can potentially lead to increased energy consumption and hardware costs due to the need for more power-hungry and costly radio frequency (RF) chains. Additionally, the introduction of numerous active components operating at high frequencies can create complex interference scenarios, pilot contamination, and significant hardware impairments.

[0005] To counter these issues, the development of cost-effective strategies for wireless communication systems is imperative and reconfigurable intelligent surfaces (RISs) are being viewed as potential solutions. Conventional reconfigurable intelligent surfaces primarily focus on surfaces that reflect incident signals. Purely reflective reconfigurable intelligent surfaces primarily function as passive relay nodes that adjust the direction of incident wireless signals, thereby improving the performance of wireless communication. However, both the source of a signal and the target of the signal need to be on the same side of the reconfigurable intelligent surface, which limits the flexibility of the reconfigurable intelligent surface. This constraint reduces the flexibility of deploying these surfaces in various real-world scenarios where the transmitter and receiver may not be located on the same side of the reconfigurable intelligent surfaces.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] In order to describe the manner in which at least some of the advantages and features of the invention may be obtained, a more particular description of embodiments of the invention will be rendered by reference to specific embodiments thereof which are illustrated in the appended drawings. Understanding that these drawings depict only typical embodiments of the invention and are not therefore to be considered to be limiting of its scope, embodiments of the invention will be described and explained with additional specificity and detail through the use of the accompanying drawings, in which:

[0007] FIG. 1 discloses aspects of a reconfigurable intelligent surface or panel;

[0008] FIG. 2 discloses aspects of a unit cell included in a metallization layer of a panel;

[0009] FIG. 3 discloses additional aspects of a panel that includes a plurality of unit cells;

[0010] FIG. 4A discloses aspects of a panel that is thermally controlled to operate in reflective and

transmission modes;

[0011] FIG. 4B discloses additional aspects of the panel in FIG. 4A;

[0012] FIG. 5 illustrates a change in resistivity for an example metal insulation transition material;

[0013] FIG. 6A discloses aspects of a panel operating in a transmission or transmissive mode;

[0014] FIG. 6B discloses aspects of a panel operating in a reflection or reflective mode;

[0015] FIG. 7A discloses aspects of the reflection mode;

[0016] FIG. 7B discloses aspects of the transmission mode;

[0017] FIG. 7C discloses aspects of a phase response for a panel;

[0018] FIG. 8 discloses aspects of controlling an operational mode of a panel; and

[0019] FIG. 9 discloses aspects of a computing device, entity, or system.

DETAILED DESCRIPTION OF SOME EXAMPLE EMBODIMENTS

[0020] Embodiments of the present invention generally relate to reconfigurable intelligent surfaces. More particularly, at least some embodiments of the invention relate to systems, hardware, software, computer-readable media, and methods for selectively reflecting or transmitting a signal incident on the reconfigurable intelligent surface.

[0021] Embodiments of the invention further relate to a reconfigurable intelligent surface that is capable of both transmitting and reflecting wireless signals. Reconfigurable intelligent surfaces in accordance with embodiments of the invention can operate as a relay node or transmission node that transmits signals to the other side of the reconfigurable intelligent surface or as a reflection node that reflects signals or electromagnetic radiation. This provides greater flexibility in deploying reconfigurable intelligent surfaces and in expanding their potential applications.

[0022] In one example, a reconfigurable intelligent surface is an engineered two-dimensional surface that may include passive elements or unit cells and may include metamaterials. A panel that includes a grid or arrangement of unit cells is an example of a reconfigurable intelligent surface. A reconfigurable intelligent surface functions by manipulating a phase response of each unit cell. This adjusts or configures the propagation of incident wireless signals and enables smart radio environments (SREs).

[0023] Embodiments of the invention more specifically relate to a reconfigurable intelligent surface that is capable of either reflecting or transmitting wireless signals (electromagnetic waves).

Reconfigurable intelligent surfaces can dynamically switch between a transmission mode and a reflection mode, thereby providing reconfigurable 360° environment coverage. When operating in the reflection mode, the reconfigurable intelligent surface behaves like an RF (Radio Frequency) mirror that reflects the incoming beam back on the same side. When operating in a transmission mode, the reconfigurable intelligent surface operates as a lens, projecting the incoming beam with high directivity on the other side. A reconfigurable intelligent surface can be deployed or installed on surfaces such as windows (walls or other locations) where the radio coverage can be selectively enhanced on both sides of the surface or window. A series of reconfigurable intelligent surfaces can be deployed and arranged to reflect a signal to a particular area. This allows obstacles that may degrade or interfere with a signal in the environment, whether indoor or outdoor, to be avoided.

[0024] FIG. 1 discloses aspects of a reconfigurable intelligent surface. FIG. 1 illustrates a reconfigurable intelligent surface **104** (or panel) that is configured to operate in a transmission mode **100** or a reflection mode **102**. In the transmission mode **102**, an incident signal **106** is transmitted through and by the reconfigurable intelligent surface **104** as the signal **108**.

Embodiments of the invention can control the direction of transmission by configuring the unit cells. In the reflection mode **102**, the incident signal **110** is reflected as the signal **112** by the reconfigurable intelligent surface **104**. The reflection direction of the reflected signal can also be controlled by configuring the unit cells. Because reconfigurable intelligent surfaces do not require radio frequency (RF) chains, they present a more economical and environmentally friendly alternative to traditional multi-antenna and relaying technologies.

[0025] Embodiments of the invention relate to a reconfigurable intelligent surface that includes a

switch layer. Controlling the switch layer allows the reconfigurable intelligent surface to switch from the transmission mode to the reflection mode and from the reflection mode to the transmission mode.

[0026] Generally, the switch layer includes a material (e.g., a metal insulator transition material or similar materials which can change their metallic properties with external stimuli) configured to transition from a metallic state to an insulator state and from the insulator state to the metallic state. Embodiments of the invention are discussed with respect to vanadium dioxide (VO.sub.2) as the switch layer. However, embodiments of the invention include other materials that have similar characteristics. These materials include VO.sub.2 and other vanadium compounds, but are not limited thereto. The transition may occur based on the temperature of the material. The temperature of the switch layer may be changed using electric energy. For example, a voltage may be applied to a strip of material that generates heat (e.g., based on its resistance). The heat generated in this manner is delivered to the switch layer to induce the desired transition.

[0027] In one example, the switch layer includes a layer of vanadium dioxide (VO.sub.2). This material, VO.sub.2, undergoes a phase change from a metallic state to an insulator state depending on a temperature of the material. The switch layer can be heated using, in one example using a heat source that is heated by applying a voltage. By controlling the voltage applied to the heat source (e.g., on or off), the reconfigurable intelligent surface can dynamically switch between a transmission mode (when VO.sub.2 is in insulator state) and a reflection mode (when VO.sub.2 is in metallic state) in real-time or near real time.

[0028] In the transmission mode, functionally comparable to an optical lens, the reconfigurable intelligent surface modifies the phase shift of different lens portions when transmitting, thereby collimating an incident wave. In the reflection mode, the reflected signal from each element/unit cell is offered a certain phase shift such that a constructive interference leads the reflected beam at the desired angle to the incoming beam.

[0029] To allow for arbitrary phases on the aperture, a minimum phase range of 360° can be provided by each unit cell. The change in reflection phase from each unit cell is achieved by making dimensional changes to the unit cell geometry. In one example, the unit cells are arranged on the reconfigurable intelligent surface panel according to the desired phase profile such that a constructive reflected signal gain is achieved in the desired direction. Embodiments of the invention offer desirable characteristics such as a broad bandwidth, low reflection loss, low insertion loss, and a compact physical profile. In a passive panel, the reflection phase of each unit cell may be determined in advance.

[0030] FIG. 2 discloses aspects of a metallic or unit cell that may be included in a reconfigurable intelligent surface. FIG. 2 illustrates an example metal pattern (**204, 206, 208**) of a unit cell **200** formed on a surface or substrate **202**. The pattern includes, by way of example only, a center portion **208** that is enclosed by a concentric rings **206** and **204**. In this example, the center portion **208** has a square shape. The concentric rings **206** and **204** also have a square shape in this example. The center portion **208** and the concentric rings **206** and **204** are generally referred to as rings or a ring pattern. The configuration of the metal pattern of each unit cell may include any shape/number of discrete components as long as the pattern **204, 206, 208** is resonant at a desired frequency (or range of frequencies) of operation in one example. Thus, the metal pattern may include a concentric configuration, a solid shape (e.g., a cross or X shape) or the like.

[0031] The example unit cell **200** of FIG. 2 describes a particular pattern and illustrates how the pattern may be varied. This example is presented by way of example and not limitation. In one example, the dimensions **210** of the outer ring **204** (S.sub.3 by S.sub.3) and the dimensions **214** of the center portion **208** (S.sub.1 by S.sub.1) are constant. The dimensions **212** of the inner or middle ring **206** (S.sub.2 by S.sub.2) in the unit cells of a panel may vary in different embodiments or in the same embodiment. Other embodiments allow for the dimensions of the center portion **208** and the rings **206** and **204** to be varied as well. More generally by way of example only, the dimensions

conform to the following condition: $S_{\text{sub.1}} \leq S_{\text{sub.2}} \leq S_{\text{sub.3}}$, for any given embodiment. In addition, the thickness **212** of the rings **204** and **206** may also vary and may be the same or different.

[0032] Generally, the center portion **208** and the rings **206** and **204** are configured (e.g., sized, shaped, and positioned) to be resonant with a particular frequency or a range of frequencies. By way of example and not limitation, embodiments of the invention relate to the transmission and reflection of signals (electromagnetic radiation) in the range of 30 GigaHertz to 300 GigaHertz. In one example, the designs/patterns of the unit cells may be frequency independent and may depend on the size of the unit cell and the spacing between the unit cells. Thus, the design of the unit cell or of the reconfigurable intelligent surface can be scaled to operate at lower frequencies or at higher frequencies including TeraHertz frequencies.

[0033] FIG. 3 discloses aspects of a reconfigurable intelligent surface. FIG. 3 illustrates a reconfigurable intelligent surface or panel **300**. The panel **300** includes a plurality of unit cells **200** formed on a substrate **203** or panel body. In this examples, each of the unit cells has the same configuration or metallic pattern. However, the configurations may vary based on the desired reflection direction or for other reasons. The metallic patterns of the unit cells **200** are not connected (e.g. electrically) to each other in one example. The overall dimensions of the panel **300** may depend on or be determined based on the desired operational frequency. The desired operational frequency, for example, may be reflected in the dimensions of the unit cells **200**.

[0034] The metallic pattern of the unit cells **200** may be formed on or printed on the substrate **302**, which may be transparent (e.g., to the relevant radio frequencies). The angle of the reflected signal or beam is determined by a phase profile of the unit cells **200**. In one example, the reflection phase of each unit cell is selected to ensure that the beam is amplified or reflected in the intended direction.

[0035] The panel **300** may be an active panel or a passive panel. In an active panel, the unit cells **200** of the panel **300** are configured to adjust a phase shift of the incoming signal, which, when managed collectively across the entire surface, can steer the signal in a specific direction or focus the reflected signal towards a particular point. This phase control may be performed using various mechanisms like PIN diodes, varactor diodes, or Micro-Electro-Mechanical Systems (MEMS) based capacitors. These mechanisms allow signal reflection in the same half space as the incoming signal. Alternatively, the panel **300** may be static and be configured in advance, (e.g., in the design of the metallic patterns of the unit cells) to reflect in a particular or specific direction.

[0036] Embodiments of the invention further relate to a panel configured to reflect and/or transmit incident signals. To achieve the flexibility of either signal reflection or transmission, the panel **300** may include a Vanadium Dioxide ($\text{VO}_{\text{sub.2}}$) layer (a switching layer) to switch between reflection and transmission modes through thermal regulation. As a material that exhibits a metal-insulator transition, $\text{VO}_{\text{sub.2}}$ can transition from a metallic to an insulating state and vice versa under the influence of heat. Consequently, the temperature of the $\text{VO}_{\text{sub.2}}$ layer can be manipulated to switch dynamically between transmission mode (when $\text{VO}_{\text{sub.2}}$ is in an insulating state) and reflection mode (when $\text{VO}_{\text{sub.2}}$ is in a metallic state), all in real time.

[0037] More specifically, $\text{VO}_{\text{sub.2}}$ undergoes a temperature-driven phase shift at about 67° C. When the temperature falls below this threshold, $\text{VO}_{\text{sub.2}}$ behaves as an insulator. Conversely, when the temperature exceeds this point, $\text{VO}_{\text{sub.2}}$ transitions into a metallic state. Each unit or element of the panel is designed to provide a full 360° phase shift through dimensional modifications. The reflection phase of each unit cell is determined by the intended direction of the panel's reflected beam.

[0038] The panel **300** may include a heating layer such that the switch layer can be heated. Tungsten (W), an example of a refractory material, may be used as a heating layer and may be configured as a meandered heater due to its high melting point, resistance to thermal stress and chemical erosion, stability under high pressure and low thermal conductivity.

[0039] In transmission mode, the panel **300** mimics a lens, guiding the incoming beam with remarkable precision to the opposite side. When in reflection mode, the panel **300** functions as an RF mirror, directing the incoming beam with high directivity back to the same side.

[0040] FIG. **4A** illustrates an example of a panel configured for to operation in reflective and transmissive modes. FIG. **4A** more specifically illustrates example layers of an example panel **400**. The panel includes a metallization layer **404** that includes metallic patterns that are formed or printed on a substrate **406**. These metallic patterns in the metallization layer **404** form the unit cells of the panel **400**. The reflection phase of each unit cell is configured to ensure that the unit cells collectively amplify the beam or reflected signal in an intended direction.

[0041] In one example, the substrate **406** is silicon or FR4. Other substrates include, but are not limited to, glass, sapphire, quartz, rogers RF or the like. In one example, the substrate includes a dielectric material.

[0042] The cover layer **402**, which may be a dielectric, is formed over the metallic pattern to protect the metallization layer **404**. A metal insulator transition layer **408** is formed on a surface of the substrate **406**. The metallization layer **404** and the metal insulator transition layer **408** are on opposite sides of the substrate **406**. The metal insulator transition layer **408** is an example of a switch layer and may be formed, by way of example, of VO.sub.2.

[0043] An insulator layer **410** is formed under the metal insulator transition layer **408**. The insulator layer **410** separates the metal insulator transition layer **408** from a refractory heater **412** or heating layer. The bottom layer **414**, which may be a dielectric, may be configured to protect the refractory heater **412**. Contact pins **416** may be provided that extend through the bottom layer **414** and connect to the refractory heater **412**. When a voltage is applied across the contact points **416**, a current flows in the refractory heater **412** and heat is generated.

[0044] In one example, the insulator layer **410** is thermally conductive such that the heat generated by the refractory heater **412** is received by the metal insulator transition layer **408**. When the appropriate temperature is reached, the metal insulator transition layer **408** transitions and the operating mode of the panel **400** changes. When the voltage is removed from the contact points **416**, heat is no longer generated by the refractory heater **412** and the metal insulator transition layer **408** cools below the transition temperature and the panel **400** switches back to the other mode.

[0045] In one example, the refractory heater **412** has a meandering shape, although other shapes including straight portions are possible. This helps ensure/control the amount of heat generated. The shape of the refractory heater **412** may be periodic repetitive, sinusoidal, or other repeating shape. The dimensions (e.g., width and thickness) of the refractory heater **412** may change and may or may not be constant. Further, sharp points in the meandering shape are avoided or not included in the shape of the refractory heater **412** to avoid heat gradients. Due to the shape of the panel, the meandering shape may have transition shapes (e.g., transition shape **422**) that allows the meandering shape to switch to a different orientation. The refractory heater **412** may have any shape, thickness, and/or width.

[0046] As is illustrated in FIG. **4A**, the refractory heater **412** includes four sides and four transitions **422**. The sides each have the same shape in this example. Ends of the meandering shape of the refractory heater **412** terminate at the contact points **416**. The specific shape can be selected based on heat requirements, transition time requirements, or the like. The specific shape may depend on panel shape. For example, a hexagon shaped panel may have a refractory heater with 6 sides.

[0047] The refractory heater **412** may also be transparent to radio frequencies (e.g., operational frequencies of the panel **400**). The refractory heater **412** may be disposed on a perimeter of the layer (e.g., a perimeter of the unit cells) or may overlap partially or completely with the unit cells in the metallization layer **404**. In this example, the refractory heater **412** and the unit cells do not overlap in the depth direction **430** (FIG. **4B**). Thus, all of the unit cells are inside a perimeter of the refractory heater **412**.

[0048] The panel **400** may include or be connected to a controller **418** and/or a power source **420**.

The controller **418** may be controlled over a wireless network and determine when a voltage is applied to the contact points **416**. Thus, the controller **418** may be configured to control when the panel **400** operates in a transmissive mode and when the panel **400** operates in a reflective mode. The controller **418** may determine when a voltage is applied/removed from the contact points **416**, thereby controlling the temperature of the switch layer **408** and the operating mode of the panel **400** [0049] FIG. **4B** illustrates a cross-sectional view of the panel **400** that is taken through the contact points. As illustrated, a refractory heater **412** is formed on a cover **414**. The insulator layer **410** is formed over the refractory heater **412** and the metal insulation transition layer is formed over the insulator layer **410**. Thus, the insulator layer **410** separates the refractory heater **412** from the metal insulation transition layer **408**. As previously stated, the insulation layer **410** is thermally conductive but not electrically conductive in one embodiment. This may prevent an electrical connection between the refractory heater **412** and the metal insulation transition layer **408**. [0050] The substrate **406** is formed over the metal insulation layer **408** and the metallization layer **404** is formed on the substrate **406**. A cover **402** is formed to protect the metallization layer **404**. The contact points **416** allow an electrical connection (e.g., a voltage or current) to be applied to the refractor heater **412**.

[0051] Different conductive metals like copper, aluminum, or gold can be used for the metallization layer **404**. The VO.sub.2 material (or other metal insulation transition material) used to form the metal insulation transition layer **408** has a transition phase that leads changes in its electrical and optical properties. At temperatures below 67° C., VO.sub.2 is in its monoclinic (insulating) phase. As the temperature rises above 67° C., VO.sub.2 undergoes a first-order phase transition to its rutile (metallic) phase. This transition leads to a change in resistivity by several orders of magnitude, from an insulator to a conductor.

[0052] FIG. **5** illustrates a change in resistivity for a metal insulation transition material such as VO.sub.2. In the graph **500**, the resistivity of the metal insulation transition material drops substantially at certain temperatures. Controlling the temperature of the metal insulation transition layer **408** thus allows the operating mode (reflective/transmissive) of the panel **400** to be controlled by, in effect, changing the resistivity of the VO.sub.2.

[0053] More specifically in one example, the phase transition of VO.sub.2 is accompanied with a hysteresis. Thus, the transition temperature when heating may be different from the transition temperature when cooling. The hysteresis of the metal insulation transition material may be considered in configuring the refractory heater **412**.

[0054] In one example, tungsten (W) is selected as the refractory heater **412** due to its high melting point of 3422° C. Tungsten is a metal with a high melting point, which makes tungsten useful for applications that involve extreme heat, such as in the electronics, aerospace, and military industries. Tungsten also has good electrical conductivity and a low coefficient of thermal expansion, making it even useful for high-temperature applications.

[0055] FIG. **6A** discloses aspects of a panel operating in a transmissive mode. In the transmissive mode, an incident signal **602** is received by a panel **602** (e.g., the panel **400**) and is transmitted as a transmitted signal **606**. The panel **600**, in this example is similar to an optical lens. The panel **600** alters the phase shift across various unit cells during transmission, which allows for the collimation of an incoming wave or signal **602**.

[0056] FIG. **6B** discloses aspects of a panel operating in a reflective or reflection mode. In the reflection mode, the metal insulator transition layer is controlled to be in a metallic state such that the incident signal **602** is reflected by the panel **604** as a reflected signal **608** or reflected beam. In this example, the signal reflected from each unit cell is given a specific phase shift, ensuring constructive interference that steers the reflected beam or signal **608** to the desired angle relative to the incoming signal **602** in one example.

[0057] FIGS. **7A** discloses aspects of the reflection mode and FIG. **7B** discloses aspects of the transmission mode. More specifically, FIGS. **7A** and **7B** illustrates aspects of the reflection and the

transmission mode of a panel for different configurations of a unit cell. FIGS. 7A and 7B illustrate curves for unit cells that correspond to dimensions S2 (see dimension **212** in FIG. 2) from 0.8 mm to 3.2 mm in increments of 0.4 mm. The curves **704** and **706** in the graph **702** of FIG. 7A correspond, respectively, to dimensions of 0.8 and 3.2 mm. The curves **714** and **716** of FIG. 7B correspond, respectively, to dimensions of 0.8 and 3.2 mm.

[0058] When the panel (e.g., the panel **400**) is configured to operate in the reflection mode, the unit cell reflects most of the signal as illustrated in the simulated magnitude of S.sub.11 (dB) at about 77 GHz as illustrated in the graph **702**. When the panel is configured to operate in the transmission mode, the simulated S.sub.21 response illustrated in the graph **712** of FIG. 7B illustrates a maximum transmission at about 77 GHz.

[0059] FIG. 7C discloses aspects of a phase response for a panel. The graph **722** in FIG. 7C illustrates a full field electromagnetic simulated variation in the full field EM reflection phase (S.sub.11 (°)) for a dimensional change (changes in dimensions S.sub.2) in the unit cell from 0.8 mm to 3.2 mm.

[0060] Embodiments of the invention thus relate to thermally controlling the configuration of a reconfigurable intelligent surface or panel to operate in a transmission or reflection mode. This allows incident signals to be either reflected or transmitted and offers selective access to both sides of the space surrounding the panel. Embodiments of the invention use a meandered heater disposed underneath or in proximity (e.g., separated by an insulator layer) a thin layer of VO.sub.2 (or other metal insulator transition material) to increase/decrease the temperature of the VO.sub.2 layer. The heater is arranged, in one example, on the periphery of the panel such that the heater pattern does not create any radio frequency interference with the unit cells in the metallization layer in one example.

[0061] Embodiments of the invention further eliminate the need for a bulky physical structure, complex feed network, or the tedious task of soldering thousands of components on-chip. The panel, in one example, is a monolithic design with a compact physical profile and a wide operational bandwidth for various frequencies or spectrums including, but not limited to, the mmWave spectrum. As far as our understanding extends, this proposal constitutes the first of its kind monolithic RIS design.

[0062] Reconfigurable intelligent surfaces can enhance the coverage area and quality of wireless networks, including in scenarios where obstructions (e.g., trees, buildings, metallic shells (cars)) impede the links between base stations or access points and end users or user equipment. These obstructions might include, by way of example, roadside trees, buildings, walls, or the metallic shells of vehicles.

[0063] For example, in outdoor communications, reconfigurable intelligent surfaces can be integrated into windows (e.g., car windows, plane windows, ship windows) to augment the signal strength using the transmission capabilities of reconfigurable intelligent surfaces discussed herein.

[0064] Embodiments of the invention also improve indoor-to-outdoor communications as well as indoor communications. In these scenarios, particularly at mmWave and THz frequencies, building walls can cause severe penetration loss, significantly limiting the coverage offered by outdoor base stations. Reconfigurable intelligent surfaces can act as a bridge from outdoor to indoor environments or to enhance coverage in indoor environments.

[0065] With indoor communications, more specifically, reconfigurable intelligent surfaces capable of both transmission and reflection are more advantageous than surfaces that only reflect signals and only provide half space coverage. Embodiments of the invention can achieve full-space coverage due to the ability of providing both transmission and reflection capabilities. This could reduce the propagation distance and consequently enhance the received signal power.

[0066] FIG. 8 discloses aspects of a method for transmitting and/or reflecting signals (e.g., electromagnetic waves). The method **800** includes deploying **802** a panel in an environment. In one example, a panel may include connections such that the panel can be connected to a power source

and/or a controller. The controller (e.g., a computing device, server, FPGA) may be local or remote with respect to the panel. The controller then controls **804** operation of the panel. The operation is controlled by placing the panel in a particular mode (e.g., transmission mode or reflection mode). This is achieved by controlling a voltage applied to a refractory heater to generate (or not generate) heat. The heat is transmitted to (or removed from) a switch layer (a metal insulation transition layer such as VO₂) that is thermally transitioned. In effect, controlling the voltage applied to the refractory heater allows the switch layer of the panel to transition from the metallic state to the insulator state or from the insulator state to the metallic state. The mode of the panel can be changed by the controller as needed, in response to user input, or for other reasons.

[0067] It is noted that embodiments of the invention, whether claimed or not, cannot be performed, practically or otherwise, in the mind of a human. Accordingly, nothing herein should be construed as teaching or suggesting that any aspect of any embodiment of the invention could or would be performed, practically or otherwise, in the mind of a human. Further, and unless explicitly indicated otherwise herein, the disclosed methods, processes, and operations, are contemplated as being implemented by computing systems that may comprise hardware and/or software. That is, such methods processes, and operations, are defined as being computer-implemented.

[0068] In general, embodiments of the invention may be implemented in connection with systems, software, and components, that individually and/or collectively implement, and/or cause the implementation of, signal processing operations, wireless coverage operations, signal steering or reflection operations, signal transmission operations, panel control operations, network coverage operations, or the like. More generally, the scope of the invention embraces any operating environment in which the disclosed concepts may be useful.

[0069] It is noted that any operation(s) of any of the methods disclosed herein may be performed in response to, as a result of, and/or, based upon, the performance of any preceding operation(s). Correspondingly, performance of one or more operations, for example, may be a predicate or trigger to subsequent performance of one or more additional operations. Thus, for example, the various operations that may make up a method may be linked together or otherwise associated with each other by way of relations such as the examples just noted. Finally, and while it is not required, the individual operations that make up the various example methods disclosed herein are, in some embodiments, performed in the specific sequence recited in those examples. In other embodiments, the individual operations that make up a disclosed method may be performed in a sequence other than the specific sequence recited.

[0070] Following are some further example embodiments of the invention. These are presented only by way of example and are not intended to limit the scope of the invention in any way.

[0071] Embodiment 1. A panel comprising: a substrate, a metallization layer formed on a first surface of the substrate, wherein the metallization layer is configured to be resonant at an operating frequency or an operating frequency range, a switch layer formed on a second surface of the substrate, wherein the switch layer is in a metallic state when a temperature of the switch layer is above a transition temperature and is in an insulator state when the temperature of the switch layer is below the transition temperature, and a refractory heater layer configured to generate heat such that the temperature of the switch layer is controlled, wherein the panel operates in a reflective state when the switch layer is in the metallic state and operates in a transmission mode when the switch layer is in the insulator state.

[0072] Embodiment 2. The panel of embodiment 1, further comprising an insulator layer positioned between the switch layer and the refractory heater layer, wherein the insulator layer is thermally conductive.

[0073] Embodiment 3. The panel of embodiment 1 and/or 2, wherein the metallization layer comprises a plurality of unit cells that are configured to have a phase profile such that the incident signal is reflected in a specific direction when operating in the reflection mode.

[0074] Embodiment 4. The panel of embodiment 1, 2, and/or 3, further comprising a cover layer

formed over the metallization layer.

[0075] Embodiment 5. The panel of embodiment 1, 2, 3, and/or 4, further comprising a bottom cover layer configured to cover the refractory heater layer.

[0076] Embodiment 6. The panel of embodiment 1, 2, 3, 4, and/or 5, further comprising contact points extending through the bottom cover layer and connecting with the refractory heater layer, wherein a voltage is applied to the refractory heater layer via the contact points.

[0077] Embodiment 7. The panel of embodiment 1, 2, 3, 4, 5, and/or 6, wherein the refractory heater layer is transparent at least to the operating frequency or range of operating frequencies.

[0078] Embodiment 8. The panel of embodiment 1, 2, 3, 4, 5, 6, and/or 7, where in the refractory heater layer comprises a strip with a meandering shape.

[0079] Embodiment 9. The panel of embodiment 1, 2, 3, 4, 5, 6, 7, and/or 8, wherein at least one of: the strip is disposed at or near a periphery of the panel to reduce radio frequency interference on the metallization layer, the meandering shape comprises transitions such that different portions of the meandering strip are oriented in different orientations, the meandering shape includes curved portions and straight portions, and the strip does not overlap with unit cells of the metallization layer in a depth direction or at least partially overlaps with the unit cells in the depth direction.

[0080] Embodiment 10. The panel of embodiment 1, 2, 3, 4, 5, 6, 7, 8, and/or 9, wherein ends of the refractory heater layer terminate at contact points.

[0081] Embodiment 11. The panel of embodiment 1, 2, 3, 4, 5, 6, 7, 8, 9, and/or 10, wherein the refractory heater layer comprises tungsten and the switch layer comprises vanadium dioxide (VO₂).

[0082] Embodiment 12. The panel of embodiment 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, and/or 11, wherein the refractory heater layer has a resistance and wherein a shape of the refractory heater layer is configured to avoid temperature gradients.

[0083] Embodiment 13. The panel of embodiment 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, and/or 12, further comprising configuring a shape, width, and thickness of the refractory heater layer.

[0084] Embodiment 14. The panel of embodiment 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, and/or 13, further comprising a controller configured to connect and disconnect a power source to the refractory heater layer.

[0085] Embodiment 15. The panel of embodiment 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, and/or 14, wherein the controller is controlled wirelessly.

[0086] Embodiment 16. The panel of embodiment 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, and/or 15, wherein the substrate comprises at least one of silicon, FR4, glass, sapphire, quartz or a dielectric material and wherein the insulator layer comprises a dielectric material and wherein the panel is transparent to radio frequencies in the transmission mode.

[0087] Embodiment 17. A system comprising: panels that are place in an environment to provide network coverage in the environment, wherein each panel comprises: a substrate, a metallization layer formed on a first surface of the substrate, wherein the metallization layer is configured to be resonant at an operating frequency or an operating frequency range, a switch layer formed on a second surface of the substrate, wherein the switch layer is in a metallic state when a temperature of the switch layer is above a transition temperature and is in an insulator state when the temperature of the switch layer is below the transition temperature, and a refractory heater layer configured to generate heat such that the temperature of the switch layer is controlled, wherein the panel operates in a reflective state when the switch layer is in the metallic state and operates in a transmission mode when the switch layer is in the insulator state.

[0088] Embodiment 18. The system of embodiment 17, wherein at least one of the panels operates in the reflective mode and at least one of the panels operates in the transmission mode.

[0089] Embodiment 19. The system of embodiment 17 and/or 18, wherein a first panel included in the panels is configured to reflect an incident signal to a second panel included in the panels.

[0090] Embodiment 20. The system of embodiment 17, 18, and/or 19, wherein each of the panels

further comprises: an insulator layer positioned between the switch layer and the refractory heater layer, wherein the insulator layer is thermally conductive, wherein the metallization layer comprises a plurality of unit cells that are configured to have a phase profile such that the incident signal is reflected in a specific direction when operating in the reflection mode, a cover layer formed over the metallization layer, a bottom cover layer formed below the refractory heater layer, and contact points extending through the bottom cover layer and connecting with the refractory heater layer, wherein a voltage is applied to the refractory heater layer via the contact points, wherein: the refractory heater layer is transparent at least to the operating frequency or range of operating frequencies, the refractory heater layer comprises a meandering strip disposed at or near a periphery of the panel to reduce radio frequency interference on the metallization layer, wherein the refractory heater layer comprises tungsten and the switch layer comprises vanadium dioxide (VO₂), wherein the refractory heater layer has a resistance and wherein a shape of the refractory heater layer is configured to avoid temperature gradients and heat the switch layer.

[0091] The embodiments disclosed herein may include the use of a special purpose or general-purpose computer including various computer hardware or software modules, as discussed in greater detail below. A computer may include a processor and computer storage media carrying instructions that, when executed by the processor and/or caused to be executed by the processor, perform any one or more of the methods disclosed herein, or any part(s) of any method disclosed.

[0092] As indicated above, embodiments within the scope of the present invention also include computer storage media, which are physical media for carrying or having computer-executable instructions or data structures stored thereon. Such computer storage media may be any available physical media that may be accessed by a general purpose or special purpose computer.

[0093] By way of example, and not limitation, such computer storage media may comprise hardware storage such as solid state disk/device (SSD), RAM, ROM, EEPROM, CD-ROM, flash memory, phase-change memory ("PCM"), or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other hardware storage devices which may be used to store program code in the form of computer-executable instructions or data structures, which may be accessed and executed by a general-purpose or special-purpose computer system to implement the disclosed functionality of the invention. Combinations of the above should also be included within the scope of computer storage media. Such media are also examples of non-transitory storage media, and non-transitory storage media also embraces cloud-based storage systems and structures, although the scope of the invention is not limited to these examples of non-transitory storage media.

[0094] Computer-executable instructions comprise, for example, instructions and data which, when executed, cause a general purpose computer, special purpose computer, or special purpose processing device to perform a certain function or group of functions. As such, some embodiments of the invention may be downloadable to one or more systems or devices, for example, from a website, mesh topology, or other source. As well, the scope of the invention embraces any hardware system or device that comprises an instance of an application that comprises the disclosed executable instructions.

[0095] Although the subject matter has been described in language specific to structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended claims is not necessarily limited to the specific features or acts described above. Rather, the specific features and acts disclosed herein are disclosed as example forms of implementing the claims.

[0096] As used herein, the term module, component, engine, agent, service, or the like may refer to software objects or routines that execute on the computing system. These may be implemented as objects or processes that execute on the computing system, for example, as separate threads. While the system and methods described herein may be implemented in software, implementations in hardware or a combination of software and hardware are also possible and contemplated. In the present disclosure, a 'computing entity' may be any computing system as previously defined

herein, or any module or combination of modules running on a computing system.

[0097] In at least some instances, a hardware processor is provided that is operable to carry out executable instructions for performing a method or process, such as the methods and processes disclosed herein. The hardware processor may or may not comprise an element of other hardware, such as the computing devices and systems disclosed herein.

[0098] In terms of computing environments, embodiments of the invention may be performed in client-server environments, whether network or local environments, or in any other suitable environment. Suitable operating environments for at least some embodiments of the invention include cloud computing environments where one or more of a client, server, or other machine may reside and operate in a cloud environment.

[0099] With reference briefly now to FIG. 9, any one or more of the entities disclosed, or implied, by the Figures and/or elsewhere herein, may take the form of, or include, or be implemented on, or hosted by, a physical computing device, one example of which is denoted at **900**. As well, where any of the aforementioned elements comprise or consist of a virtual machine (VM), that VM may constitute a virtualization of any combination of the physical components disclosed in FIG. 9.

[0100] In the example of FIG. 9, the physical computing device **900** includes a memory **902** which may include one, some, or all, of random access memory (RAM), non-volatile memory (NVM) **904** such as NVRAM for example, read-only memory (ROM), and persistent memory, one or more hardware processors **906**, non-transitory storage media **908**, UI device **910**, and data storage **912**. One or more of the memory components **902** of the physical computing device **900** may take the form of solid state device (SSD) storage. As well, one or more applications **914** may be provided that comprise instructions executable by one or more hardware processors **906** to perform any of the operations, or portions thereof, disclosed herein.

[0101] Such executable instructions may take various forms including, for example, instructions executable to perform any method or portion thereof disclosed herein, and/or executable by/at any of a storage site, whether on-premises at an enterprise, or a cloud computing site, client, datacenter, data protection site including a cloud storage site, or backup server, to perform any of the functions disclosed herein. As well, such instructions may be executable to perform any of the other operations and methods, and any portions thereof, disclosed herein.

[0102] The present invention may be embodied in other specific forms without departing from its spirit or essential characteristics. The described embodiments are to be considered in all respects only as illustrative and not restrictive. The scope of the invention is, therefore, indicated by the appended claims rather than by the foregoing description. All changes which come within the meaning and range of equivalency of the claims are to be embraced within their scope.

Claims

1. A panel comprising: a substrate; a metallization layer formed on a first surface of the substrate, wherein the metallization layer is configured to be resonant at an operating frequency or an operating frequency range; a switch layer formed on a second surface of the substrate, wherein the switch layer is in a metallic state when a temperature of the switch layer is above a transition temperature and is in an insulator state when the temperature of the switch layer is below the transition temperature; and a refractory heater layer configured to generate heat such that the temperature of the switch layer is controlled, wherein the panel operates in a reflective state when the switch layer is in the metallic state and operates in a transmission mode when the switch layer is in the insulator state.
2. The panel of claim 1, further comprising an insulator layer positioned between the switch layer and the refractory heater layer, wherein the insulator layer is thermally conductive.
3. The panel of claim 1, wherein the metallization layer comprises a plurality of unit cells that are configured to have a phase profile such that the incident signal is reflected in a specific direction

when operating in the reflection mode.

4. The panel of claim 1, further comprising a cover layer formed over the metallization layer.

5. The panel of claim 1, further comprising a bottom cover layer configured to cover the refractory heater layer.

6. The panel of claim 1, further comprising contact points extending through the bottom cover layer and connecting with the refractory heater layer, wherein a voltage is applied to the refractory heater layer via the contact points.

7. The panel of claim 1, wherein the refractory heater layer is transparent at least to the operating frequency or range of operating frequencies.

8. The panel of claim 1, where in the refractory heater layer comprises a strip with a meandering shape.

9. The panel of claim 8, wherein at least one of: the strip is disposed at or near a periphery of the panel to reduce radio frequency interference on the metallization layer; the meandering shape comprises transitions such that different portions of the meandering strip are oriented in different orientations; the meandering shape includes curved portions and straight portions; and the strip does not overlap with unit cells of the metallization layer in a depth direction or at least partially overlaps with the unit cells in the depth direction.

10. The panel of claim 1, wherein ends of the refractory heater layer terminate at contact points.

11. The panel of claim 1, wherein the refractory heater layer comprises tungsten and the switch layer comprises vanadium dioxide (VO₂).

12. The panel of claim 1, wherein the refractory heater layer has a resistance and wherein a shape of the refractory heater layer is configured to avoid temperature gradients.

13. The panel of claim 1, further comprising configuring a shape, width, and thickness of the refractory heater layer.

14. The panel of claim 1, further comprising a controller configured to connect and disconnect a power source to the refractory heater layer.

15. The panel of claim 14, wherein the controller is controlled wirelessly.

16. The panel of claim 1, wherein the substrate comprises at least one of silicon, FR4, glass, sapphire, quartz or a dielectric material and wherein the insulator layer comprises a dielectric material and wherein the panel is transparent to radio frequencies in the transmission mode..

17. A system comprising: panels that are place in an environment to provide network coverage in the environment, wherein each panel comprises: a substrate; a metallization layer formed on a first surface of the substrate, wherein the metallization layer is configured to be resonant at an operating frequency or an operating frequency range; a switch layer formed on a second surface of the substrate, wherein the switch layer is in a metallic state when a temperature of the switch layer is above a transition temperature and is in an insulator state when the temperature of the switch layer is below the transition temperature; and a refractory heater layer configured to generate heat such that the temperature of the switch layer is controlled, wherein the panel operates in a reflective state when the switch layer is in the metallic state and operates in a transmission mode when the switch layer is in the insulator state.

18. The system of claim 17, wherein at least one of the panels operates in the reflective mode and at least one of the panels operates in the transmission mode.

19. The system of claim 17, wherein a first panel included in the panels is configured to reflect an incident signal to a second panel included in the panels.

20. The system of claim 17, wherein each of the panels further comprises: an insulator layer positioned between the switch layer and the refractory heater layer, wherein the insulator layer is thermally conductive, wherein the metallization layer comprises a plurality of unit cells that are configured to have a phase profile such that the incident signal is reflected in a specific direction when operating in the reflection mode; a cover layer formed over the metallization layer; a bottom cover layer formed below the refractory heater layer; and contact points extending through the

bottom cover layer and connecting with the refractory heater layer, wherein a voltage is applied to the refractory heater layer via the contact points, wherein: the refractory heater layer is transparent at least to the operating frequency or range of operating frequencies, the refractory heater layer comprises a meandering strip disposed at or near a periphery of the panel to reduce radio frequency interference on the metallization layer, wherein the refractory heater layer comprises tungsten and the switch layer comprises vanadium dioxide (VO_2), wherein the refractory heater layer has a resistance and wherein a shape of the refractory heater layer is configured to avoid temperature gradients and heat the switch layer.
